

A Gift of Fire

Third edition

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Chapter 9: Professional Ethics and Responsibilities

What We Will Cover

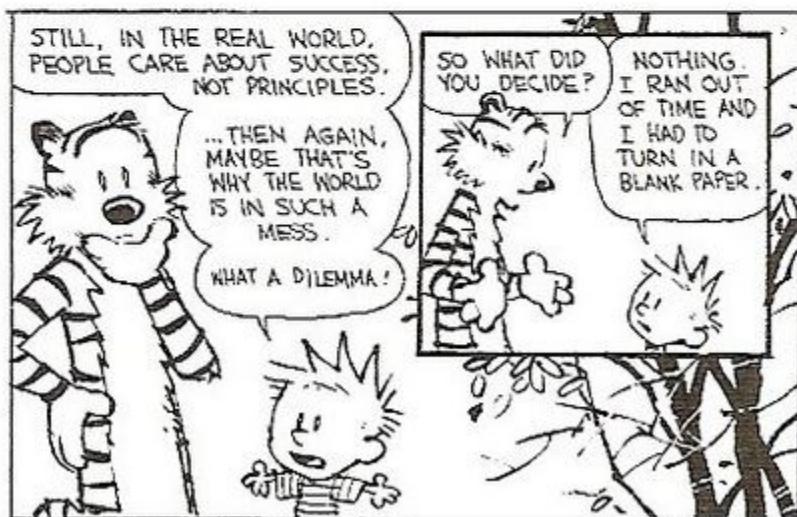
- What is Professional Ethics?
- Ethical Guidelines for Computer Professionals
- Scenarios

calvin and hobbes



THEN I THOUGHT, LOOK, CHEATING ON ONE LITTLE TEST ISN'T SUCH A BIG DEAL. IT DOESN'T HURT ANYONE.

...BUT THEN I WONDERED IF I WAS JUST RATIONALIZING MY UNWILLINGNESS TO ACCEPT THE CONSEQUENCE OF NOT STUDYING.



“Calvin: Today at school, I tried to decide whether to cheat on my test or not.

Calvin: I wondered, is it better to do the right thing and fail...or is it better to do the wrong thing and succeed?

Calvin: On the one hand, undeserved success gives no satisfaction. ... But on the other hand, well-deserved failure gives no satisfaction either.

Calvin: Of course, most everybody cheats some time or other. People always bend the rules if they think they can get away with it. ... Then again, that doesn't justify *my* cheating.

Calvin: Then I thought, look, cheating on one little test isn't such a big deal. It doesn't hurt anyone. ... But then I wondered if I was just rationalizing my unwillingness to accept the consequence of not studying.

Calvin: Still, in the real world, people care about success, not

Calvin: Still, in the real world, people care about success, not principles. ... Then again, maybe that's why the world is in such a mess. What a dilemma!

Hobbes: So what did you decide?

Calvin: Nothing. I ran out of time and I had to turn in a blank paper.

Hobbes: Anymore, simply acknowledging the issue is a moral victory.

Calvin: Well, it just seemed wrong to cheat on an ethics test.

What is "Professional Ethics"?

- Professional ethics includes relationships with and responsibilities toward customers, clients, coworkers, employees, employers, others who use one's products and services, and others whom they affect
- A professional has a responsibility to act ethically. Many professions have a code of ethics that professionals are expected to abide by
 - Medical doctors
 - Lawyers and judges
 - Accountants

What is "Professional Ethics"? (cont.)

- There are special aspects to making ethical decisions in a professional context
- Honesty is one of the most fundamental ethical values; however, many ethical problems are more subtle than the choice of being honest or dishonest
- Some ethical issues are controversial

Some Important Distinctions

- Right, wrong, and okay
- Negative and Positive Rights
- Distinguishing Wrong and Harm
- Separating goals from constraints
- Personal preference and ethics
- Law and Ethics

Ethical Guidelines for Computer Professionals

Guidelines and Professional Responsibilities:

- Understand what success means
- Include users (such as medical staff, technicians, pilots, office workers) in the design and testing stages to provide safe and useful systems
- Do a thorough, careful job when planning and scheduling a project and when writing bids or contracts
- Design for real users

Ethical Guidelines for Computer . . . (cont.)

Guidelines and Professional Responsibilities (cont.):

- Don't assume existing software is safe or correct; review and test it
- Be open and honest about capabilities, safety, and limitations of software
- Require a convincing case for safety
- Pay attention to defaults
- Develop communication skills

Scenarios

Methodology:

- Brainstorming phase:
 - List all the people and organizations affected (the stakeholders)
 - List risks, issues, problems, and consequences
 - List benefits. Identify who gets each benefit
 - In cases where there is no simple yes or no decision, but rather one has to choose some action, list possible actions

Scenarios (cont.)

Methodology:

- Analysis phase
 - Identify responsibilities of the decision maker
 - Identify rights of stakeholders
 - Consider the impact of the options on the stakeholders (consequences, risks, benefits, harms, costs)
 - Categorize each potential action as ethically obligatory, prohibited, or acceptable
 - When there are multiple options, select one, considering the ethical merits of each, courtesy to others, practicality, self-interest, personal preferences, etc.

Reinforcing Exclusion

- Voice Recognition
- Hand writing
- Nontechnical users
- Different ethnic groups
- Disabled people
- Older people
- Children

Copying an employee's file

- You are a computer system manager. An employee is out sick and another employee requests that you copy all files from the sick person's computer to his so he can do some work.

Scenarios

Scenario 1:

- Your team is working on a computer controlled laser device for treating cancerous tumors. The computer controls direction, intensity, and timing of the beam that destroys the tumor. Various delays have put the project behind schedule, and the deadline is approaching. There will not be time to complete all the planned testing. The system has been functioning properly in the routine treatment scenario that have been tested so far. You are the project manager, and you are considering whether to deliver the system on time, while continuing testing, and to make patches if bugs are found.

Scenarios

Scenario 2:

- Suppose you are a member of a team working on a computer-controlled crash avoidance system for automobiles. You think the system has a flaw that could endanger people. The project manager does not seem concerned and expects to announce completion of the project soon. Do you have an ethical obligation to do something?

Scenarios

Scenario 3:

- Your company has 25 licenses for a computer program, but you discover that it has been copied onto 80 computers.

Scenarios

Scenario 5:

- You work for the IRS, the Social Security Administration, a movie-rental company, or an Internet service provider. Someone asks you to get a copy of records about a particular person. He will pay you \$500.

Scenarios

Scenario 6:

- You have a small consulting business. The CyberStuff company plans to buy software to run a new collaborative content-sharing Web site. CyberStuff wants to hire you to evaluate bids from vendors. Your spouse works for NetWorkx and did most of the work in writing the bid that NetWorkx plans to submit. You read the bid while your spouse was working on it and you think it is excellent. Do you tell CyberStuff about your spouse's connection with NetWorkx?

Scenarios

Scenario 7:

- A team of programmers is developing a communications system for firefighters to use when fighting a fire. Firefighters will be able to communicate with each other, with supervisors near the scene, and with other emergency personnel. The programmers will test the system in a field near the company office.

Scenarios

Scenario 8:

- You are the computer system administrator for a mid-sized company. You can monitor the company network from home, and you frequently work from home. Your niece, a college student, is visiting for a week. She asks to use your computer to check her e-mail. Sure, you say. You are being a gracious host. What is the ethical problem?

Scenarios

Scenario 9:

- You are a relatively junior programmer working on modules that collect data from loan application forms and convert them to formats required by the parts of the program that evaluate the applications. You find that some demographic data are missing from some forms, particularly race and age. What should your program do? What should you do?